



AMFU-Net: A multi-scale attention network for lesion segmentation in fused breast ultrasound imaging

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ARTICLE INFO

Keywords:

Breast ultrasound tomosynthesis
Image fusion
AMFU-net
Deep learning
Semantic segmentation

ABSTRACT

Breast lesions represent a serious threat to women's health, highlighting the need for accurate and efficient diagnostic imaging techniques. Since no single imaging modality can fully characterize breast tissue, ultrasound computed tomography (USCT) combines reflection and transmission imaging to provide complementary structural and acoustic information. This paper proposes a novel fusion strategy based on the HIS discrete wavelet transform, optimized via the Particle Swarm Optimization (PSO) algorithm, to integrate reflection and transmission images into a fused breast ultrasound imaging containing both modalities' salient features. To further improve lesion segmentation performance, we introduce AMFU-Net—an Attention-guided Multi-scale Feature Extraction and Fusion Network. AMFU-Net incorporates a Multi-scale Feature Extraction and Fusion (MFEF) module at the bottleneck layer and a Dual-branch Attention Gate (DAG) in the encoder-decoder skip connections, enabling adaptive weighting and enhanced cross-scale contextual feature integration. Experimental evaluations were performed on a clinical breast USCT dataset consisting of 215 annotated slices from 210 patients, which was partitioned into training, validation, and test sets in an 8:1:1 ratio. AMFU-Net achieved a Dice coefficient of 91.42%, outperforming the classical U-Net by 2.36%. This work provides a comprehensive solution that synergistically combines multimodal image fusion and advanced deep segmentation, offering significant potential for accurate breast lesion diagnosis and clinical decision support.

1. Introduction

According to recent statistics from the American Cancer Society, approximately 319,750 new breast cancer cases and 42,680 related deaths are projected in the United States for 2025 [1]. The increasing prevalence of this public health issue underscores the critical importance of improving and innovating breast cancer diagnostic imaging technologies. Ultrasound Computed Tomography (USCT) has become an important imaging tool for breast tumor screening, as it realizes multi-parameter image reconstruction of tissue structure, sound velocity distribution, and attenuation coefficient through simultaneous acquisition of reflected signal and transmitted data [2]. As the core of computer-aided diagnosis (CAD) systems, the accurate segmentation of breast lesions directly affects the effectiveness of clinical diagnosis and treatment planning.

However, in the USCT technique, the reflected signal mainly characterizes the morphological features of anatomical structures. In contrast, the transmitted data allow quantitative characterization of the acoustic property parameters of the tissue [3–7]. Notably, there are inherent

limitations of a single imaging modality in clinical applications: reflection imaging is insufficiently sensitive to acoustic parameters, while transmission imaging is relatively limited in terms of tissue structural information [8,9]. Also, in clinical applications, USCT images tend to have low resolution and blurred lesion boundaries, so segmenting breast lesions remains a challenging task.

Early medical image segmentation often relied on conventional pipelines, such as thresholding, edge-based, region-based, and clustering-based techniques, which typically depend on hand-crafted intensity/texture cues and heuristic post-processing. Recent surveys have summarized these classical strategies and their limitations, especially when dealing with low-resolution images, low signal-to-noise ratio, and fuzzy lesion boundaries (e.g., ultrasound-related artifacts). [10–12] In contrast, convolutional neural networks (CNNs) have enabled end-to-end learning of hierarchical features and have become the dominant paradigm in medical image segmentation, consistently improving accuracy and robustness across diverse clinical scenarios. [10,

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[11] Representative architectures, including U-Net [13], CA-Net [14], HyperDenseNet [15], and other deep CNN-based tumor segmentation networks [16,17], have demonstrated superior performance compared with conventional approaches, as also reported in recent reviews. [10, 11] A symmetric encoder–decoder structure and a skip-connection algorithm were used to create the U-Net architecture that Ronneberger [13] suggested. Connections (skip connections), which use minimal medical data to accomplish end-to-end pixel-level segmentation. Yap et al. employed three FCN variants, including U-Net, to segment breast lesions and demonstrated their superiority over conventional methods [18]. Eventually, as deep learning continued to advance, additional segmentation network designs seemed to be used for breast segmentation. The attention mechanism and multi-scale information extraction and fusion are the two primary focuses of these techniques. Multiscale dilation convolution was employed by Shareef [19] and Qian [20] to efficiently extract multiscale feature information from breast lesions and enhance segmentation accuracy. By integrating the attention mechanism with the encoder and decoder, a number of researchers have improved the segmentation performance of complex cases [21–23]. Additionally, the dynamic feature weighting mechanism has improved the feature representation of the tumor's fine structure.

Although numerous studies have investigated CNN-based segmentation for MRI, CT, and conventional ultrasound, our survey indicates that several critical challenges remain unaddressed in the context of breast USCT. First, existing segmentation architectures are not tailored to the unique characteristics of USCT images, which are marked by low contrast, indistinct lesion boundaries, and substantial variability in lesion size. Second, prior multimodal imaging research seldom incorporates adaptive fusion mechanisms; most fusion strategies rely on manually selected or fixed fusion rules, thereby limiting flexibility and generalizability. Third, limited attention has been given to the interplay between the fusion stage and the segmentation stage, resulting in suboptimal utilization of the complementary structural and acoustic information inherent in USCT.

To address these gaps, the key contributions of this study are summarized as follows:

1. We propose an adaptive multimodal fusion framework based on Discrete Wavelet Transform and an improved Particle Swarm Optimization algorithm.

Unlike traditional fusion methods that employ fixed or manually defined fusion rules, the proposed approach automatically optimizes fusion coefficients to integrate complementary structural (reflection-mode) and acoustic (transmission-mode) information. This adaptive fusion mechanism alleviates the inherent limitations of single-modality USCT imaging and enhances overall information representation.

2. We develop AMFU-Net, a segmentation network designed for breast USCT images, incorporating a Dual-branch Attention Gate (DAG) and a Multi-scale Feature Extraction and Fusion (MFEF) module.

The DAG modules applied to skip connections (layers 1–4) facilitate boundary-aware and context-enhanced feature refinement, while the MFEF module at the bottleneck captures rich multi-scale semantic representations. Together, these components effectively address common USCT challenges such as low resolution, blurred lesion boundaries, and significant lesion size variation.

3. We conduct extensive experiments on a clinical breast USCT dataset to evaluate the proposed fusion–segmentation pipeline.

Experimental results demonstrate that our method delivers superior lesion segmentation performance compared with mainstream segmentation networks, confirming the efficacy of integrating adaptive multimodal fusion with attention-guided multi-scale learning for accurate breast lesion analysis.

2. Related work

2.1. Attention mechanism

The integration of attention mechanisms has significantly advanced medical image segmentation by enabling networks to focus on diagnostically relevant regions while suppressing irrelevant background information.

Early progress in this direction was marked by the work of Oktay et al. [24], who introduced spatial attention gates (AGs) into the skip connections of the U-Net architecture. These gates leverage coarse contextual cues from the decoder to modulate encoder features, enabling the network to emphasize foreground structures and refine segmentation boundaries. Following this, channel attention mechanisms gained increasing attention. Lee et al. [21] proposed replacing the traditional global average pooling (GAP) operation in channel attention with multiscale grid average pooling (MSGAP). MSGAP fuses local details with global context via dynamic grid partitioning, enhancing the accuracy of breast cancer ultrasound image segmentation.

Subsequent research explored hybrid attention models that jointly consider spatial and channel dependencies. Woo et al. [25] introduced the Convolutional Block Attention Module (CBAM), which sequentially applies channel attention — using both GAP and Global Max Pooling (GMP) — followed by spatial attention, derived from channel-aggregated features. This lightweight yet effective mechanism enhances the network's ability to focus on informative regions across both dimensions. Roy et al. [26] proposed the scSE module (spatial and channel squeeze & excitation), which conducts spatial and channel recalibration in parallel, offering complementary feature refinement. Iqbal and Sharif [27] presented MDA-Net for breast ultrasound segmentation, combining a multi-scale fusion (MF) module with a dual-attention mechanism that integrates channel attention (cA) and lesion-specific attention (lA), enabling more robust delineation of lesion boundaries.

More recently, the success of self-attention mechanisms and Transformer architectures in natural language processing has inspired their adoption in medical imaging. Wang et al. [28] introduced TransUNet, a hybrid architecture that integrates Transformer-based encoders — capable of capturing long-range dependencies — with CNN-based decoders for precise localization. Building on this idea, Cao et al. [29] proposed SwinUNet, which is entirely constructed using Swin Transformer blocks. These blocks utilize shifted windows to efficiently compute self-attention while preserving hierarchical representations, enabling effective feature modeling across scales.

Despite notable progress, prior attention mechanisms still exhibit several limitations. Most early models, such as Attention U-Net [24] and CBAM [25], focus predominantly on either local spatial or global channel relationships, often neglecting the integration of both. Their fixed-scale designs struggle to adapt to lesions of diverse sizes and morphological complexity. Moreover, many hybrid attention modules introduce excessive computational overhead due to redundant channel-spatial recalibration, making real-time clinical deployment difficult. Transformer-based models [28,29], while capable of modeling long-range dependencies, demand large-scale annotated data and heavy computation, which limits their practical utility in medical settings.

2.2. Multi-scale feature extraction and fusion

Architectural innovations like Unet++ [30] improved feature propagation through nested dense connections between the encoder and decoder, enabling multi-scale fusion via enhanced skip connections. Chen et al. [31] proposed atrous spatial pyramid pooling (ASPP), which employs parallel dilated convolutional layers to capture multi-scale context while mitigating gradient diffusion. Cai et al. [32] demonstrated that deeper placement of ASPP modules in U-Net significantly boosts segmentation accuracy. In CA-Net [14], the multi-scale context fusion (MCF) module integrates spatial context fusion (SCF) for

Table 1
Summary of attention mechanisms and multi-scale feature fusion methods in medical image segmentation.

Method	Technical characteristics	Contribution	Limitations
Attention mechanisms			
Oktay et al. (Attention U-Net) [24]	Spatial Attention Gate (AG) in skip connections	Highlights relevant regions, suppresses irrelevant background	Local context only; fixed receptive fields; global semantic information is weak
Lee et al. (MSGRAP) [21]	Multi-scale grid average pooling replaces GAP in channel attention	Captures fine-grained features by combining local and global information	Static pooling design; lacks adaptive response to variable lesion size
Woo et al. (CBAM) [25]	Sequential channel + spatial attention using GAP and GMP	Lightweight; enhances both dimensional representations	Sequential design limits cross-dimension synergy; coarse spatial resolution
Roy et al. (scSE) [26]	Parallel spatial and channel recalibration	Combines complementary attentions; improves feature refinement	Ignores inter-channel dependencies; spatial modeling lacks depth
Iqbal et al. (MDA-Net) [27]	Dual attention: channel + lesion-specific, with multi-scale fusion	Enhances lesion boundary and robustness	Lesion-specific attention depends on prior knowledge; complex and heavy
Wang et al. (TransUNet) [28]	Transformer encoder + CNN decoder	Captures long-range dependencies; enhances global context modeling	Requires large datasets; high computational and memory cost
Cao et al. (SwinUNet) [29]	Pure Swin Transformer with shifted window attention	Maintains hierarchy; efficient self-attention over local-global scales	High complexity; inference slow; limited for edge-aware tasks
Multi-scale feature extraction and fusion			
Zhou et al. (UNet++) [30]	Nested dense skip connections	Enhances feature reuse and multiscale learning	Topological redundancy; high memory and training cost
Chen et al. (ASPP) [31]	Parallel dilated convolutions with varying rates	Captures multiscale context; reduces gradient vanishing	Sensitive to kernel size; may miss fine boundary details
Cai et al. (ASPP deep-level integration) [32]	ASPP added at encoder bottleneck	Improves deeper semantic capture	Fixed module depth; not adaptive across datasets
Gu et al. (CA-Net) [14]	Multi-Scale Context Fusion (MCF) + Channel attention	Aggregates hierarchical features; improves discrimination	Lacks cross-scale dynamic weighting; spatial fusion is static

multi-level feature aggregation and channel attention for dimension calibration. However, current fusion mechanisms lack dynamic cross-dimensional weight allocation, limiting their discriminative power for multi-scale pathological features. Additionally, densely nested structures risk topological redundancy and high computational overhead (see Table 1).

3. Proposed methods

Both the segmentation network architecture and the image fusion network architecture are introduced in this section, along with the distinct functions of each module within the segmentation network structure. We build a HIS (Hue-Intensity-Saturation) discrete wavelet transform (DWT) fusion framework for the image fusion network based on the particle swarm optimization (PSO) algorithm. This multi-objective dynamic optimization mechanism enables the adaptive fusion of cross-modal features. Specifically, the transmitted image is first converted to the HIS color space, and then the intensity components of the reflected and transmitted images are decomposed at multiple scales to generate low-frequency approximation subbands and high-frequency detail subbands. For the low-frequency component, a nonlinear weighted fusion strategy based on improved particle swarm optimization is introduced. The mean, information entropy, and variance are used to construct a multi-objective fitness function, and dynamic inertia weights are employed to adjust the particle search range and iteratively obtain the optimal fusion coefficients. For the high-frequency subbands, the maximum value selection method is adopted. This fusion strategy enhances structural details in the reflection image while preserving acoustic parameters (e.g., sound velocity) from the transmission image. For the image segmentation network, we add a dual-branch attention gate in the skip connection to optimize the information selection from encoder to decoder. Additionally, a multi-scale feature extraction and fusion (MFEF) module is inserted at the bottleneck, combining dilated convolutions and feature pyramid aggregation to prioritize discriminative features for lesions of varying scales.

In this study, Eqs. (1)–(8) correspond to standard formulations widely used in HIS color space transformation, DWT-based decomposition and reconstruction, and conventional PSO updating rules. These equations follow established definitions commonly adopted in the literature. In contrast, Eqs. (9)–(22) are specifically developed for this work. They introduce the customized fusion weight optimization strategy, the enhanced PSO fitness function, and the application-specific fusion rules that enable the adaptive integration of reflection and transmission USCT images.

3.1. Discrete Wavelet Transform based on Particle Swarm Optimization Algorithm

We propose an image fusion framework based on Discrete Wavelet Transform and Particle Swarm Optimization, aiming to integrate the complementary information from ultrasound reflection and transmission images. DWT is widely used in medical image fusion due to its ability to decompose images at multiple levels and scales, effectively separating low-frequency structural information and high-frequency detail features [33,34]. However, traditional DWT-based fusion methods often rely on fixed weighting schemes (e.g., averaging or maximum selection), which lack adaptability in balancing the contributions from different modalities. To address this, we introduce the Particle Swarm Optimization algorithm to dynamically optimize fusion weights. PSO, as an efficient global optimization method, has been successfully applied to multi-objective image fusion tasks, enabling iterative search for optimal fusion coefficients to enhance the entropy and visual quality of fused images [35]. By combining the multi-scale decomposition capability of DWT with the adaptive optimization mechanism of PSO, our method effectively fuses structural details from reflection images and acoustic parameters from transmission images, thereby providing more discriminative input for subsequent segmentation tasks.

3.1.1. HIS fusion technique

In image processing, RGB and HIS are two popular color representation models. The HIS model is based on the idea of human visual perception and is more accurate in representing the subjective color

perception of the human eye. The model breaks down color information into three separate parts: hue (H) indicates the type of color (red, green, blue, etc.); saturation (S) indicates the degree of color mixing with white, or the purity of the color; and intensity (I) indicates the image's brightness information, which indicates the amount of light that the human eye can perceive [33].

Two significant features of the HIS model are as follows: (1) its three components are independent of one another, meaning that when processing the luminance (I component), the hue and saturation will not be impacted, improving the image's structural details while maintaining the color information; and (2) the H and S components are more in line with the subjective color perception of the human eye, making the model especially appropriate for medical image processing and multimodal image fusion. These features also help to improve color fidelity and vision. This approach is especially well-suited for multimodal picture fusion and medical image processing, which enhances visual consistency and color fidelity.

In the HIS coding structure, the I component usually carries more information and has a higher bit representation requirement, so in the image fusion process, this component is prioritized for structural and texture feature extraction and fusion. In contrast, the H and S components remain unchanged and are involved in the color reconstruction of the final image. Based on the above considerations, the fusion strategy based on the HIS model is adopted in this study, where the feature fusion operation is performed for the I component, while the H and S components are kept stable, so as to enhance the structural and detailed information of the image, and at the same time, ensure the consistency of the color perception and the overall visual quality of the image.

A general HIS based is explained by the following equations:

$$\begin{pmatrix} I \\ v_1 \\ v_2 \end{pmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 \end{bmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}, \quad (1)$$

$$H = \tan^{-1} \left(\frac{v_1}{v_2} \right), \quad (2)$$

$$S = \sqrt{v_1^2 + v_2^2}. \quad (3)$$

where v_1 and v_2 are the transitional values in the above equations.

The inverse transform is:

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \end{bmatrix} \begin{pmatrix} I \\ S \sin(H) \\ S \cos(H) \end{pmatrix} \quad (4)$$

3.1.2. Discrete wavelet transform

DWT is an effective multi-resolution analysis tool widely used in the fields of image processing and image fusion. By filtering and downsampling each row and column of the input image in turn, the method decomposes the image into four non-overlapping subbands: a low-frequency approximation subband (LL) and three directional high-frequency detail subbands (LH, HL, and HH), which correspond to the horizontal, vertical, and diagonal details, respectively. The size of each subband image is half of the original image, reflecting the image structure characteristics at different scales.

In the wavelet decomposition process, the LL subband mainly contains the low-frequency information of the image, i.e., its main structural and luminance features, and is usually considered as the energy concentration region of the image. Therefore, this subband can be further decomposed iteratively to obtain a deeper multi-scale approximate representation. The high-frequency subbands LH, HL, and HH reflect the edge and texture details of the image in different directions, respectively, and have high spatial resolution [34].

Based on multiresolution analysis theory, the DWT achieves hierarchical image decomposition and reconstruction through filter banks.

The mathematical formulation of the two-dimensional DWT is presented below.

For a high-resolution image c_{j_1+1} , the decomposition process is expressed as:

$$\begin{cases} c_{j_1}(k_1, m_1) = \sum_{l_1, n_1} h_{l_1-2k_1} h_{n_1-2m_1} c_{j_1+1}(l_1, n_1) \\ d_{j_1}^{(1)}(k_1, m_1) = \sum_{l_1, n_1} h_{l_1-2k_1} g_{n_1-2m_1} c_{j_1+1}(l_1, n_1) \\ d_{j_1}^{(2)}(k_1, m_1) = \sum_{l_1, n_1} g_{l_1-2k_1} h_{n_1-2m_1} c_{j_1+1}(l_1, n_1) \\ d_{j_1}^{(3)}(k_1, m_1) = \sum_{l_1, n_1} g_{l_1-2k_1} g_{n_1-2m_1} c_{j_1+1}(l_1, n_1) \end{cases} \quad (5)$$

The reconstruction formula to recover the original image from low-resolution components is:

$$\begin{aligned} c_{j_1+1}(k_1, m_1) = & \sum_{l_1, n_1} \bar{h}_{k_1-2l_1} \bar{h}_{m_1-2n_1} c_{j_1}(l_1, n_1) \\ & + \sum_{l_1, n_1} \bar{h}_{k_1-2l_1} \bar{g}_{m_1-2n_1} d_{j_1}^{(1)}(l_1, n_1) \\ & + \sum_{l_1, n_1} \bar{g}_{k_1-2l_1} \bar{h}_{m_1-2n_1} d_{j_1}^{(2)}(l_1, n_1) \\ & + \sum_{l_1, n_1} \bar{g}_{k_1-2l_1} \bar{g}_{m_1-2n_1} d_{j_1}^{(3)}(l_1, n_1) \end{aligned} \quad (6)$$

where c_{j_1} is the approximation function on low resolution (e.g., LL). $d_{j_1}^{(1)}$, $d_{j_1}^{(2)}$ and $d_{j_1}^{(3)}$ are three wavelet coefficients containing local spatial details (e.g., HH, HL, and LH). c_{j_1+1} denotes the image coefficients of the high-resolution raw image. k_1 and m_1 denote the rows and columns on the scale j_1 , respectively, and the distributions of l_1 and n_1 denote the rows and columns on $j_1 + 1$. h is a low-pass decomposition filter, g is a high-pass decomposition filter, $\bar{h}$ is a low-pass reconstruction filter, and $\bar{g}$ is a high-pass reconstruction filter.

3.1.3. Particle Swarm Optimization algorithm

Particle Swarm Optimization (PSO) is a meta-heuristic method that can be used to optimize continuous nonlinear functions. It was inspired by the social behavior of schools of fish, flocks of birds, and other groupings of species. As an example, consider the flocking of birds. In PSO, each optimization problem's solution resembles a bird in the search space, which is referred to as a "particle". Each particle has two characteristic quantities: position and velocity. Each particle updates its own velocity and position based on its current position $\mathbf{x}_i(t)$, its current velocity $\mathbf{v}_i(t)$, its individual historical optimal position $\mathbf{p}_i$; and the population historical optimal position $\mathbf{g}$.

Velocity Update:

$$\mathbf{v}_i(t+1) = \omega \cdot \mathbf{v}_i(t) + c_1 r_1 \cdot (\mathbf{p}_i - \mathbf{x}_i(t)) + c_2 r_2 \cdot (\mathbf{g} - \mathbf{x}_i(t)) \quad (7)$$

where ω is the inertia weight, controlling the influence of historical velocity; c_1 and c_2 are the cognitive learning factor and social learning factor, respectively; r_1 and r_2 are random numbers within the range [0, 1], introducing randomness.

Position Update:

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \quad (8)$$

Particles continuously adjust their flight paths by tracking their individual best solution $\mathbf{p}_i$ and the global best solution $\mathbf{g}$, ultimately driving the entire swarm to converge towards the optimal solution region of the problem.

3.2. Fusion architecture

The proposed fusion framework, depicted in Fig. 1, processes the reflection image and the I-component (Intensity) of the transmission image through the following steps:

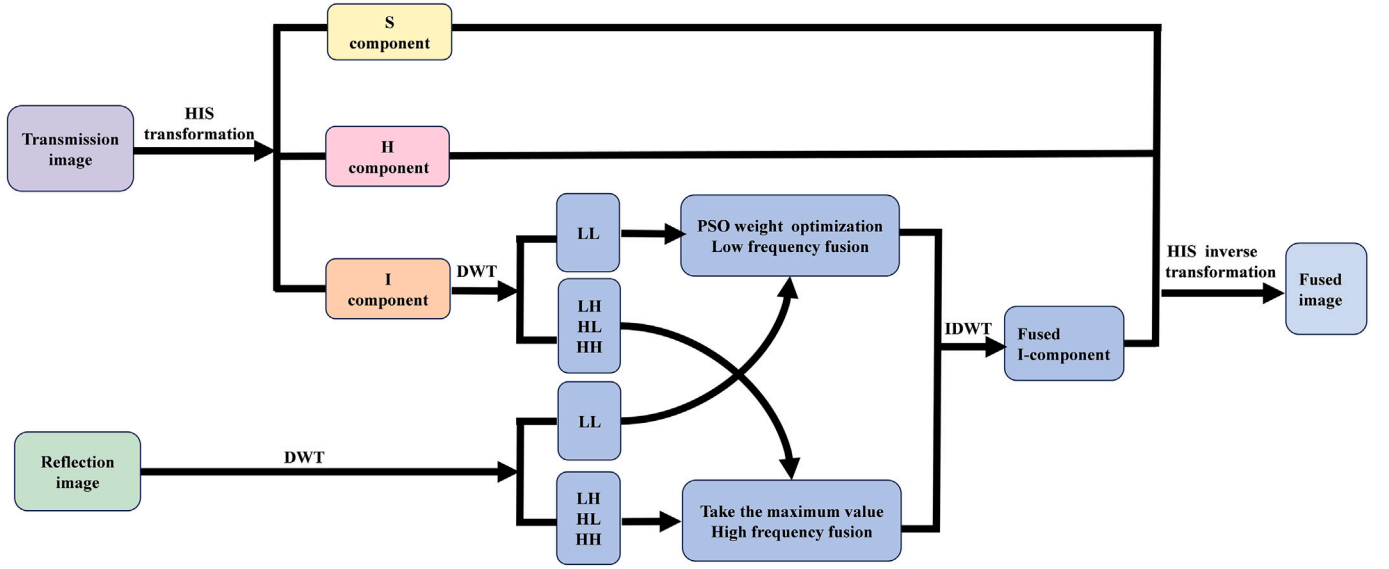


Fig. 1. Fusion network architecture with inputs as reflected image and transmitted image and output as a fused image of both.

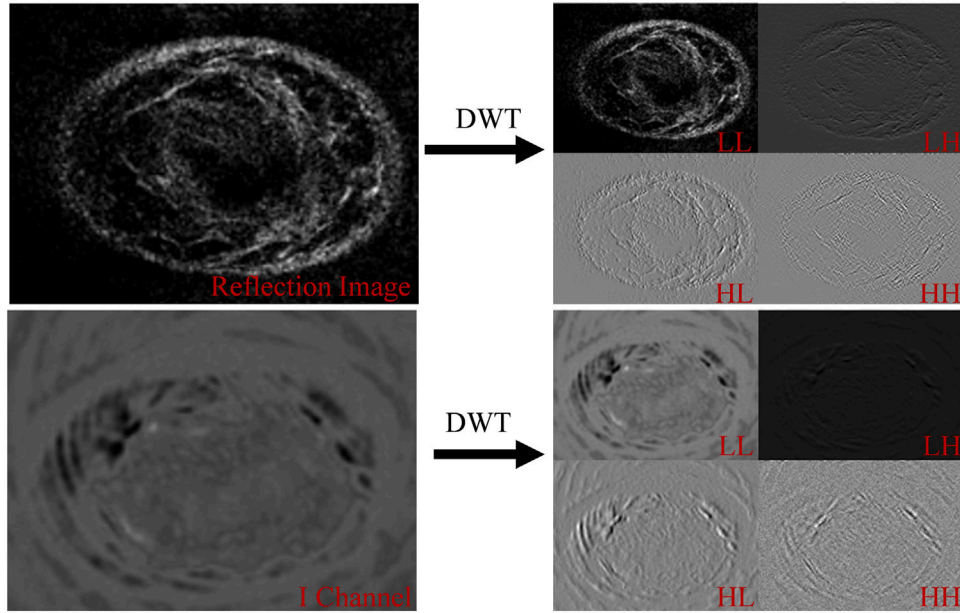


Fig. 2. The figure illustrates the low-frequency approximation (LL) and high-frequency detail (LH, HL, HH) subband images obtained from the I-channel images of the original reflection and transmission images decomposed by the first-stage discrete wavelet transform, which reveals the distribution of the frequency information in different directions.

1. Color Space Conversion:

The transmission image is converted from RGB to HIS (Hue-Intensity-Saturation) space, separating the luminance component (I) from chrominance components (H, S) to enable independent optimization of structural and parametric features.

2. Multi-scale Decomposition:

The reflection image and the I-component of the transmission image are decomposed via Discrete Wavelet Transform (DWT) into four subbands: LL (Low-Low): Preserves low-frequency information (e.g., acoustic parameters like sound velocity). LH (Low-High), HL (High-Low), HH (High-High): Capture high-frequency edge and texture details. The subband decomposition of the reflected image and the I channel of the transmitted image is shown in Fig. 2

3. Adaptive Fusion Strategies:

The main goal of the proposed method is to enhance the overall visual quality of the fused image by improving contrast while preserving

structural integrity. To this end, the image fusion process is formulated as a multi-objective optimization problem. The particle swarm optimization (PSO) algorithm is employed, relying on two key components: (1) the solution space, which represents the set of weighting parameters for each fusion stage; and (2) the fitness function, an objective metric used to evaluate the quality of the fused image.

The effectiveness of PSO heavily depends on the design of the fitness function. In this study, a multi-objective fitness function in the form of a product is adopted to simultaneously optimize contrast and information retention:

$$C = \left(\frac{\sigma^2}{\mu} \right) \times (H_{\text{before}} - H_{\text{after}}), \quad (9)$$

where σ^2 and μ denote the variance and mean of the fused (or enhanced) image, H_{before} is the entropy of the input image, and H_{after} is the entropy of the output image.

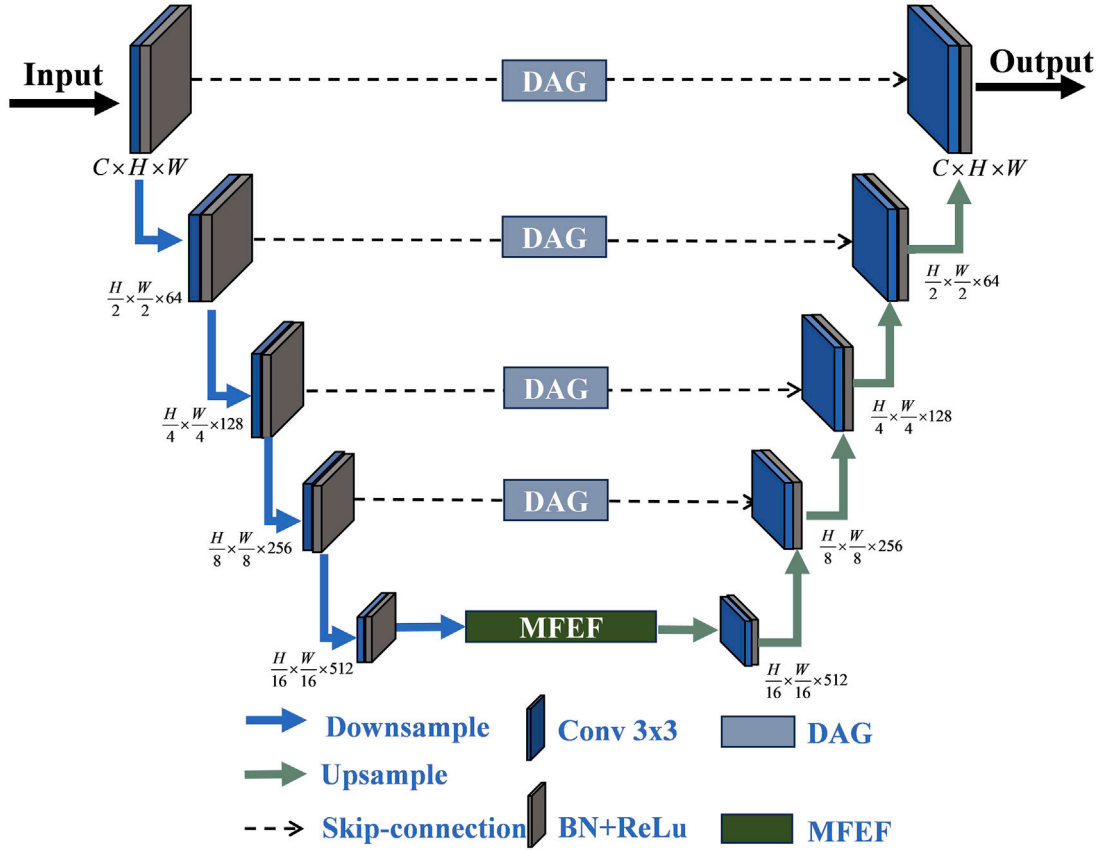


Fig. 3. Overview of the proposed AMFU-Net. The architecture improves on the U-Net network by integrating a dual-branch attention gate at the Layers 1–4 skip connections and incorporating an MFEF module at the bottleneck of the network.

All fusion and enhancement weights are automatically determined via PSO without human intervention, enabling good adaptability and generalization across multimodal image fusion tasks.

(1) Low-Frequency Fusion

In the wavelet domain, the LL subbands of the reflection image and the transmission image's I-component contain the essential structural and intensity information. To effectively combine the complementary information from both, their LL subbands are fused using a weighted average, as defined in Eq. (10):

$$LL_{\text{fusion}} = \theta_1 \times LL_{\text{reflect}} + (1 - \theta_1) \times LL_{\text{trans}}, \quad (10)$$

Here, θ_1 is the weight parameter optimized by the improved PSO algorithm. To ensure a balanced trade-off between texture detail preservation, contrast enhancement, and structural fidelity, the following fitness function is used:

$$C = \left(\frac{\sigma_{\text{fusion}}^2}{\mu_{\text{fusion}}} \right) \times (H_{\text{reflect}} - H_{\text{fusion}}), \quad (11)$$

where σ_{fusion}^2 and μ_{fusion} represent the variance and mean of the fused LL subband, H_{reflect} is the entropy of the reflection image, and H_{fusion} is the entropy of the fused subband. The PSO algorithm iteratively searches the solution space to determine the optimal θ_1 , yielding a fusion result that balances structural clarity and information content.

(2) High-Frequency Fusion The high-frequency subbands (LH, HL, HH), which primarily capture edge and texture information, are fused using a pixel-wise maximum selection strategy:

$$LH_{\text{fusion}} = \max(LH_{\text{reflect}}, LH_{\text{trans}}). \quad (12)$$

This approach effectively preserves sharp edges and fine details by avoiding the blurring effect common in weighted averaging, thereby improving the visual clarity of the fused image.

(3) Secondary Enhancement of the Fused I-Component

To further improve contrast and perceptual clarity, a secondary enhancement is applied to the fused I-component. This enhancement integrates three classic techniques: Histogram Equalization (HE), Canny Edge Detection (CED), and Median Filtering (MF), as shown in Eq. (13):

$$I_{\text{enhanced}} = \theta_2[0] \cdot HE + \theta_2[1] \cdot CED + (1 - \theta_2[0] - \theta_2[1]) \cdot MF. \quad (13)$$

Here, the weights $\theta_2[0]$ and $\theta_2[1]$ are also optimized via PSO. The corresponding fitness function is:

$$C = \left(\frac{\sigma_{\text{enhanced}}^2}{\mu_{\text{enhanced}}} \right) \times (H_{\text{fusion}} - H_{\text{enhanced}}), \quad (14)$$

where $\sigma_{\text{enhanced}}^2$ and μ_{enhanced} are the variance and mean of the enhanced image, while H_{fusion} and H_{enhanced} denote the entropy before and after enhancement, respectively.

This adaptive strategy enables dynamic adjustment of enhancement ratios according to image characteristics, thereby improving contrast while suppressing noise and enhancing perceptual quality.

3.3. AMFU-Net network architecture

Fig. 3 displays the AMFU-Net network design. An enhanced network built on top of U-net is called AMFU-net.

The encoder and decoder modules are implemented by the architecture using max-pooling and 3×3 convolutions. In order to optimize the information from the encoder to the decoder and dynamically modify the feature weights, a dual-branch attention gate (DAG) is integrated into the skip connections to capture both local and global information. As a result, encoder features are more efficiently used to produce more accurate predicted segmentation maps, and more robust features

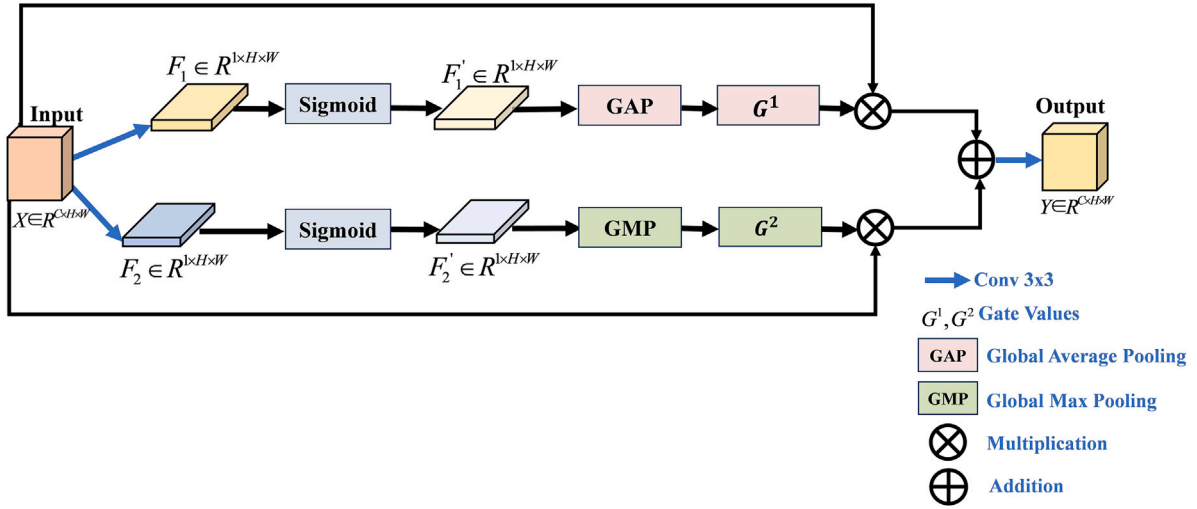


Fig. 4. Overview of the dual-branch attention gate. This module adopts a dual-branch structure combining global average and max pooling to generate attention gates, which recalibrate input features through weighted fusion, followed by convolution to enhance key feature extraction and cross-layer integration.

are recovered from the lesion region during segmentation. A multi-scale feature extraction and fusion module (MFEF) is integrated at the network bottleneck, inspired by Cai et al. [32] for hierarchical feature aggregation.

Specifically, feature maps from different encoder levels are first spatially aligned via upsampling or downsampling. These multi-level features, capturing shallow appearance patterns and deep semantic cues, are connected and passed to the MFEF module. In MFEF, features are processed through parallel convolutional branches with kernel sizes of 1×1 , 3×3 , and 5×5 to extract representations of different receptive fields. A 3×3 dilated convolution (rate = 2) is used to further capture the global context. Meanwhile, the Global Average Pool (GAP) is used to summarize the global feature statistics. These outputs are fused and refined through a spatial attention mechanism, where a 1×1 convolution following Softmax generates attention weights to emphasize informative regions and suppress irrelevant noise. This hierarchical aggregation strategy allows the network to integrate multi-level and multi-scale features at bottlenecks, effectively combining spatial detail and semantic richness.

3.4. Dual-branch attention gate

Fig. 4 shows the Dual-branch Attention Gate (DAG) in action. In medical image segmentation, most existing attention mechanisms in segmentation networks tend to focus on local features while neglecting global contextual information, often at the cost of high computational complexity. To address these limitations, we propose a dual-branch attention gate integrated into skip connections, which synergizes global max pooling and global average pooling to enhance feature selection between encoder and decoder. This mechanism dynamically balances critical feature preservation and computational efficiency.

The proposed module processes a 3D convolutional feature map $x \in \mathbb{R}^{C \times H \times W}$ through two parallel branches. First, 3×3 convolutions generate two distinct feature representations F_1 and F_2 , capturing complementary spatial-channel characteristics. Both branches apply sigmoid activation to normalize feature responses within $[0, 1]$, yielding calibrated features F'_1 and F'_2 .

Global average pooling on F'_1 produces channel attention gates G^1 , encoding contextual statistics across channels. Concurrently, global max pooling on F'_2 generates spatial attention gates G^2 highlighting salient regional patterns. These gating signals adaptively recalibrate the original input features through element-wise multiplication and summation:

$$G^1 = \text{GAP}(\sigma(\text{Conv}(x))). \quad (15)$$

$$G^2 = \text{GMP}(\sigma(\text{Conv}(x))). \quad (16)$$

$$Y = \text{Conv}(x \odot G^1 + x \odot G^2). \quad (17)$$

where σ denotes sigmoid activation, $\odot$ represents element-wise multiplication, GAP indicates global average pooling, and GMP indicates global max pooling. This architecture effectively suppresses non-informative feature propagation while enhancing cross-layer semantic integration, ultimately improving segmentation accuracy for complex medical imaging tasks.

3.5. Multi-scale Feature Extraction and Fusion Module (MFEF)

The Multi-scale Feature Extraction and Fusion Module (MFEF) shown in Fig. 5 integrates adaptive multiscale feature weighting to enhance the model's sensitivity to critical patterns. In order to capture local details, mid-range context, and long-term dependencies, the module uses parallel convolutions with varying kernel sizes (1×1 , 3×3 , 5×5) in two coordinated feature extraction and fusion routes. Based on our ablation tests and prior empirical data [36], this design was carefully selected.

Specifically, 1×1 convolution was used to preserve pixel-level identity information and efficiently reduce channel dimensionality. 3×3 convolution is widely recognized as the standard setting for contextual feature extraction, providing a good balance between localization and computational efficiency. 5×5 convolution, while more computationally intensive, captures a larger background window and facilitates the identification of larger lesion structures. To complement these, global average pooling (GAP) provides overall global semantic information, which is particularly important in medical image segmentation where foreground-background contrast is subtle. 3×3 dilated convolution (rate = 2) further expands the receptive field without increasing the kernel size, allowing the model to capture a wider range of contextual dependencies, which is crucial for recognizing ambiguous lesion boundaries.

A strong depiction that strikes a balance between local detail preservation and global context awareness is produced by combining these scales. Ablation tests were used to validate the design (see Table 5). These studies demonstrated that the Dice and other metrics performed worse when either of these branches was deleted, validating their complementary roles in the module.

The input features are first processed through parallel 1×1 , 3×3 , and 5×5 convolutional layers and concatenated with GAP-upsampled

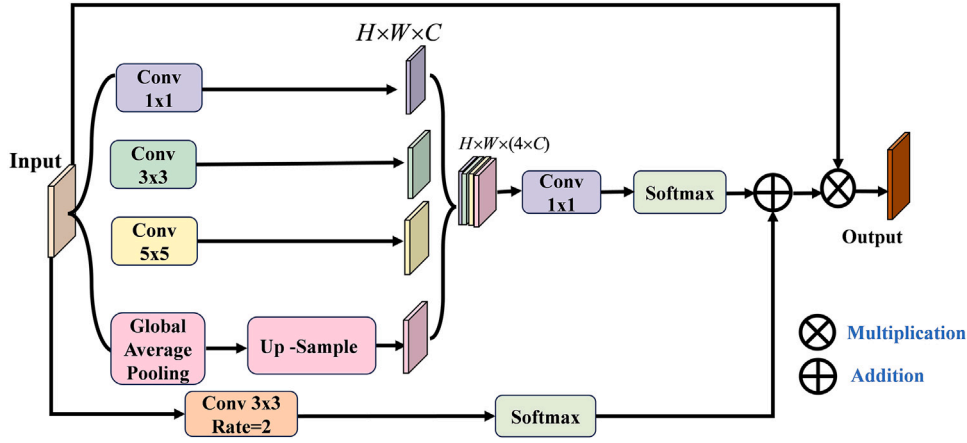


Fig. 5. Overview of the multi-scale feature extraction and fusion module. This module illustrates a multi-branch attention module that aggregates multi-scale features using convolutions of different kernel sizes and global context, followed by a softmax-based function to refine the input feature map.

features to form a multi-scale representation. A 1×1 convolution compresses the concatenated features into a unified tensor M_1 followed by ReLU activation and Softmax normalization to generate channel attention weights ω_1 .

$$M_1 = \text{Conv}_{1 \times 1}(\text{Concat}(C_{1 \times 1}, C_{3 \times 3}, C_{5 \times 5}, \text{GAP})) \quad (18)$$

$$\omega_1 = \text{Softmax}(M_1) \quad (19)$$

Simultaneously, the input undergoes dilated convolution (3×3 , rate = 2) and ReLU activation to produce spatial feature maps M_2 , which are normalized via Softmax to yield spatial attention weights ω_2 .

$$M_2 = \text{ReLU}(\text{Conv}_{3 \times 3, \text{rate}=2}(F_{\text{input}})) \quad (20)$$

$$\omega_2 = \text{Softmax}(M_2) \quad (21)$$

The final adaptive weights ω are computed as the average of ω_1 and ω_2 , dynamically recalibrating the input features to emphasize discriminative regions. The output preserves spatial resolution while integrating multi-scale context, aligning with the dual-path structure shown in the diagram.

$$\omega = 0.5 * \omega_1 + 0.5 * \omega_2 \quad (22)$$

4. Experiments

4.1. Data preparation

Clinical data were prospectively collected from January 2023 to December 2024 using the 512-element ring-array Breast Ultrasound Computed Tomography system at several medical institutions, including Beijing Tiantan Hospital affiliated with Capital Medical University and the General Hospital of the Chinese People's Liberation Army. A total of 210 female patients aged between 24 and 46 years, all previously diagnosed with breast lesions via magnetic resonance imaging (MRI), were enrolled in this study.

During the imaging process, each subject was placed in a supine position while the Breast Ultrasound Computed Tomography system automatically acquired cross-sectional images of the breast in a top-to-bottom scanning mode. The breast was immersed in a coupling medium and enclosed by a circular ultrasonic transducer array, enabling high-resolution tomographic imaging. The overall data acquisition workflow is illustrated in Fig. 6.

All acquired image sequences were annotated using the open-source tool Labelme, under the supervision of experienced radiologists. Lesion areas were labeled with a pixel value of 1, and the background

was assigned a value of 0, forming binary segmentation masks. After preprocessing and quality control, a total of 215 representative 2D tomographic slices with a resolution of 256×256 pixels were retained. The dataset was randomly split into training, validation, and test sets with a ratio of 8:1:1. To improve model generalization and mitigate overfitting, common data augmentation techniques — such as translation, horizontal flipping, random rotation, and scaling — were applied to the training data.

To analyze the potential impact of dataset size on segmentation performance, we performed a controlled data-scaling experiment under the same 8:1:1 split. Specifically, the validation set (10%) and test set (10%) were kept fixed, and only the training set size was varied. From the full training pool (80% of 215 slices), we randomly sampled 20%, 40%, 60%, 70%, and 80% of the total dataset for training (i.e., 43, 86, 129, 151, and 172 slices), while using the fixed validation set for model selection and the fixed test set for final evaluation. For each subset size, the model was trained using identical settings and evaluated on the same test set in terms of Dice (and other metrics) to characterize the performance trend and identify potential saturation.

4.2. Experimental settings

All experiments were conducted using Python 3.12 on an NVIDIA RTX 3090 GPU. The proposed AWFU-Net was trained using the Adam optimizer with an initial learning rate of 0.001. To stabilize training and accelerate convergence, a learning rate scheduler was employed to reduce the learning rate by half every 10 epochs. To prevent overfitting, L2 regularization (weight decay) was applied with a coefficient of $1e-5$. The model was trained for 200 epochs with a batch size of 12.

4.3. Clinical validation protocol

To ensure that the proposed framework was evaluated under clinically relevant conditions, a formal clinical validation protocol was established. The study was conducted under approval from the Institutional Review Board (IRB) of the participating hospitals, and written informed consent was obtained from all enrolled patients. The dataset consisted exclusively of real clinical cases collected prospectively from two medical centers using a 512-element ring-array USCT system.

For clinical validation, the entire dataset was divided into independent training, validation, and test sets, where the test set contained patient cases that were never seen during model development. This independent test set served as the basis for assessing the clinical performance of the segmentation model. All ground-truth lesion masks were annotated by two board-certified radiologists with more than



Fig. 6. Clinical data collection system.

10 years of experience in breast imaging, and disagreements were resolved through consensus reading.

To further evaluate clinical usability, segmentation outputs on the test set were reviewed by an external radiologist who was not involved in the annotation process. The reviewer assessed whether the predicted boundaries were acceptable for clinical interpretation, with a focus on lesion visibility, completeness of contour, and potential diagnostic utility. This procedure ensured that the evaluation reflected not only numerical performance but also clinical interpretability.

4.4. Evaluation metrics

4.4.1. Image fusion

To quantitatively evaluate the performance of the image fusion module, three widely adopted metrics were used:

Information Entropy (EN): Reflects the amount of information contained in the image. Higher entropy indicates richer structural and texture details.

Average Gradient (AG): Captures image sharpness and edge clarity.

Standard Deviation (SD): Measures the dispersion of pixel intensities, with larger values indicating better contrast.

These metrics were computed on the fused images to assess the preservation of complementary features and perceptual quality.

4.4.2. Image segmentation

We employed four widely used assessment metrics to quantitatively evaluate the efficacy of lesion segmentation: the average surface distance (ASD) [37], the 95% Hausdorff distance (95HD) [38], the Jaccard index (Jaccard) [39], and the Dice similarity coefficient (Dice) [40]. The following formula is used to determine these metrics:

$$Dice(A, B) = \frac{2 \times |A \cap B|}{|A| + |B|} \quad (23)$$

$$Jaccard(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (24)$$

$$95HD(A, B) = \max \left\{ \sup_{x \in A} \inf_{y \in B} \|x - y\|, \sup_{y \in B} \inf_{x \in A} \|x - y\| \right\} \quad (25)$$

$$ASD(A, B) = \frac{\sum_{x \in A} \min_{y \in B} d(x, y)}{|A|} \quad (26)$$

where A and B are the segmentation result and the true value, and x and y are the pixels in A and B , respectively.

5. Results

In this section, we present a comprehensive evaluation of the proposed AMFU-Net framework from multiple perspectives, including image fusion effectiveness, segmentation performance, module contribution, interpretability, robustness, and computational efficiency. First, we examine the fusion quality between transmission and reflection images, visually and quantitatively demonstrating how complementary modalities enhance structural clarity and tissue visibility. We then compare the segmentation performance of AMFU-Net with several state-of-the-art methods under consistent training conditions, highlighting its superiority in terms of accuracy and boundary delineation.

To further dissect the architecture, we perform ablation studies on the Dual-branch Attention Gate (DAG) and the Multi-scale Feature Extraction and Fusion (MFEF) modules, revealing their individual and combined contributions to performance improvement. Visualization analyzes — including attention maps and Grad-CAM — are employed to enhance the transparency and interpretability of the model's feature learning mechanisms. Additionally, we explore how lesion-specific attributes such as size, contrast, and shape influence segmentation outcomes, confirming the model's robustness and adaptability in handling diverse clinical cases. Finally, we assess the computational overhead associated with the PSO-based fusion and segmentation process, verifying the feasibility of applying the framework in time-sensitive clinical scenarios.

5.1. Image fusion

The fusion results shown in Fig. 7 illustrate the effective integration of acoustic parameters from the transmission image and structural details from the reflection image. By complementary fusion of the reflection image with clear contours and rich internal tissue structures and the transmission image containing functional tissue parameters, the fused image exhibits both structural clarity and enhanced tissue visibility.

To further evaluate the fusion quality at different stages, Fig. 8 presents the DWT decomposition and inverse HIS reconstruction results of the initially fused I-channel. As shown in the left part of the figure, the LL, LH, HL, and HH subbands obtained through single-level DWT illustrate the spatial distribution of structural and detail information. The right part shows the reconstructed image produced by applying the

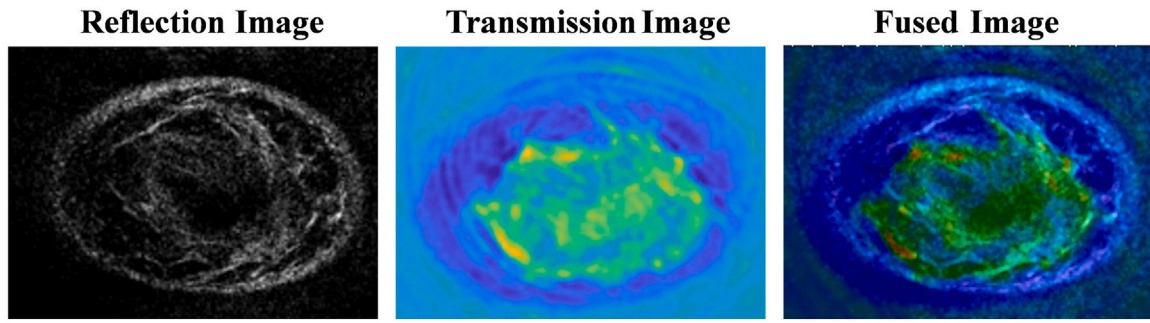


Fig. 7. Fusion result. By complementary fusion of reflectance images with clear contours and rich internal tissue details with transmission images containing tissue feature parameters, fusion images with clear contours and visualization of functional information are generated.

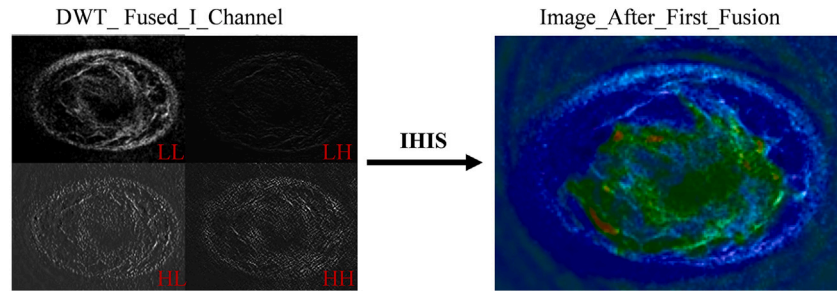


Fig. 8. Illustration of the fusion process. Left: DWT decomposition of the initially fused I-channel, showing details in the LL, LH, HL, and HH subbands. Right: The final fused image was reconstructed using the fused I-channel and the original H and S components through an inverse HIS transform.

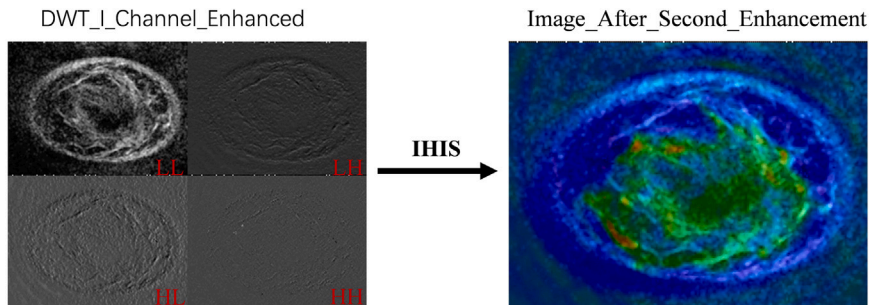


Fig. 9. Visualization after secondary enhancement. Left: DWT decomposition of the enhanced I-channel, showing the improved distribution of structural and detail features across the subbands. Right: Final image reconstructed by inverse HIS transform using the enhanced I-channel and the original H and S components.

inverse HIS transform using the fused I-channel along with the original H and S components.

In addition, Fig. 9 visualizes the results after the secondary enhancement. The left side of the figure depicts the DWT subbands of the enhanced I-channel, where improved contrast and sharpened textures are evident. The right side shows the final reconstructed image based on the enhanced I-channel and original chromatic components via inverse HIS. These results further verify that the proposed method not only retains structural fidelity but also significantly improves perceptual quality through adaptive enhancement.

As shown in Table 2, the fused image achieves a higher entropy (7.97 vs. 7.85/5.81) and an average gradient (18.07 vs. 4.44/8.66) compared to individual modalities, indicating enriched information and enhanced edge expression. Although its standard deviation (68.72) is lower than the transmission image (78.46), it significantly surpasses the reflection image (32.64), suggesting a balanced contrast without over-enhancement.

To further illustrate the quantitative improvement achieved through the proposed DWT-PSO fusion strategy, we additionally plotted bar charts of the three fusion quality metrics, including entropy (EN), average gradient (AG), and standard deviation (SD). As shown in

Table 2

Comparison of data before and after image fusion.

Metrics	Reflection image	Transmission image	Fused image
Entropy	5.81	7.85	7.97
Average gradient	8.66	4.44	18.07
Standard deviation	32.64	78.46	68.72

Fig. 10, the fused image consistently achieves the highest EN and AG values, indicating richer structural information and sharper edge representation compared to the reflection-only and transmission-only inputs. Although the SD value of the fused image is slightly lower than that of the transmission image, it remains substantially higher than that of the reflection image, confirming that the fusion process achieves a desirable balance between contrast enhancement and noise suppression.

5.2. Effect of fusion on segmentation performance

To quantify the contribution of the proposed fusion framework to the segmentation task, we conducted an experiment comparing three

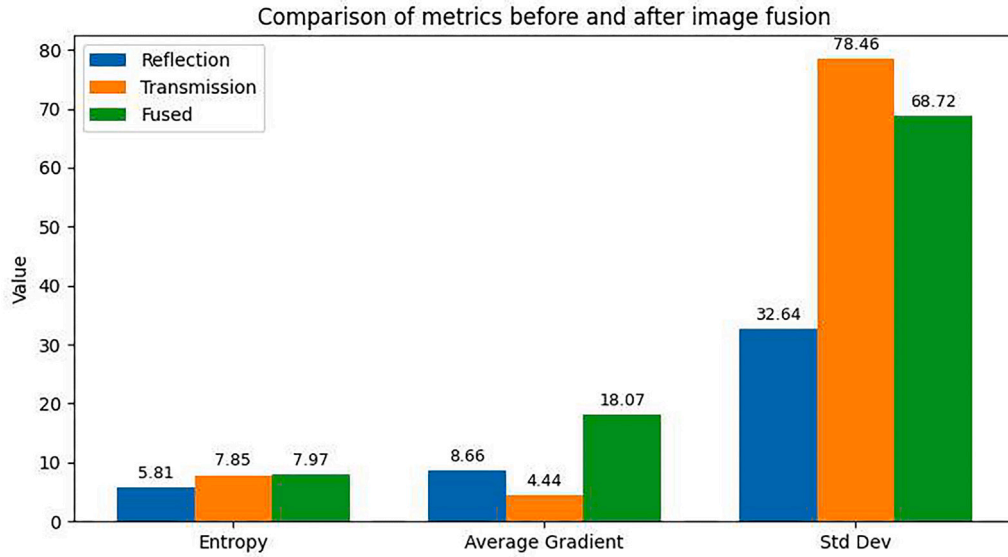


Fig. 10. The bar chart compares three widely used indicators—entropy (EN), average gradient (AG), and standard deviation (SD). The fused image shows the highest EN and AG values, demonstrating improved information richness and edge sharpness, while maintaining balanced contrast as reflected by the SD measure.



Fig. 11. Qualitative comparison between non-fused and fused inputs for the segmentation task. From left to right: reflection image, transmission image, fused image, ground-truth mask, segmentation result using only the reflection image, segmentation result using only the transmission image, and segmentation result using the fused image.

Table 3
Segmentation performance on single-modality and fused images.

Input modality	Dice (%)	Jaccard (%)	95HD	ASD
Reflection image only	85.67 ± 5.33	76.89 ± 6.01	10.12 ± 1.45	3.45 ± 0.67
Transmission image only	87.32 ± 4.21	78.45 ± 5.12	11.34 ± 1.78	4.12 ± 0.89
Fused image	91.42 ± 3.43	84.54 ± 3.98	5.61 ± 1.59	1.36 ± 0.79

input settings: (1) reflection image only, (2) transmission image only, and (3) fused image generated by the DWT-PSO framework.

All settings were evaluated using the same AMFU-Net architecture and identical training/validation/test splits to ensure fairness. The results in Table 3 demonstrate a clear improvement when using the fused image as input.

The results indicate that when using fused images as input, AMFU-Net significantly outperforms scenarios utilizing single-modality images across all evaluation metrics. Notably, in terms of the Dice coefficient, the fused images achieved an improvement of approximately 5.75% compared to reflection-only images and approximately 4.1% compared to transmission-only images. This validates that the proposed fusion strategy effectively integrates structural information from reflection images and acoustic parameters from transmission images, providing the segmentation network with more discriminative feature representations and thereby substantially enhancing segmentation accuracy and boundary localization capability.

As shown in Fig. 11, the model fed with the fused image produces a segmentation mask that best matches the ground truth, accurately capturing the shape and extent of the target region. In contrast, the segmentations obtained from the reflection-only and transmission-only images either miss part of the structure or introduce spurious responses.

These visual results confirm that fusing reflection and transmission images provides more complete and discriminative information, leading to improved segmentation performance compared with non-fused inputs.

5.3. Ablation experiments

To investigate the individual contributions of each module in AMFU-Net, we conducted ablation experiments by incrementally integrating the Dual-branch Attention Gate (DAG) and the Multi-scale Feature Extraction and Fusion (MFEF) module into the U-Net baseline. The results are summarized in Table 4.

The DAG module effectively filters and transfers salient features from the encoder to the decoder using dual-branch attention, thereby improving segmentation performance. The MFEF module enables the extraction of multiscale lesion features and strengthens edge information, enhancing both stability and boundary accuracy. The combined architecture significantly outperforms the baseline, demonstrating the synergistic benefits of both modules.

To further confirm each block's efficacy, we conducted a visual interpretability study in addition to the numerical enhancements. In particular, we contrasted the attention mappings of the skipped connections between the first and third layers of the proposed AMFU-Net

Table 4
Comparison of the functionality of each module.

Methods	Dice (%)	Jaccard (%)	95HD	ASD
<i>U-Net</i>	89.06 $\pm$ 5.12	80.85 $\pm$ 6.12	9.24 $\pm$ 1.13	3.03 $\pm$ 0.52
<i>U-Net + DAG</i>	90.34 $\pm$ 3.56	82.81 $\pm$ 3.88	9.09 $\pm$ 0.96	2.89 $\pm$ 0.41
<i>U-Net + MFEF</i>	90.19 $\pm$ 3.71	82.82 $\pm$ 4.16	7.38 $\pm$ 0.84	1.74 $\pm$ 0.53
U-Net+DAG+MFEF	91.42 $\pm$ 3.43	84.54 $\pm$ 3.98	5.61 $\pm$ 1.59	1.36 $\pm$ 0.79

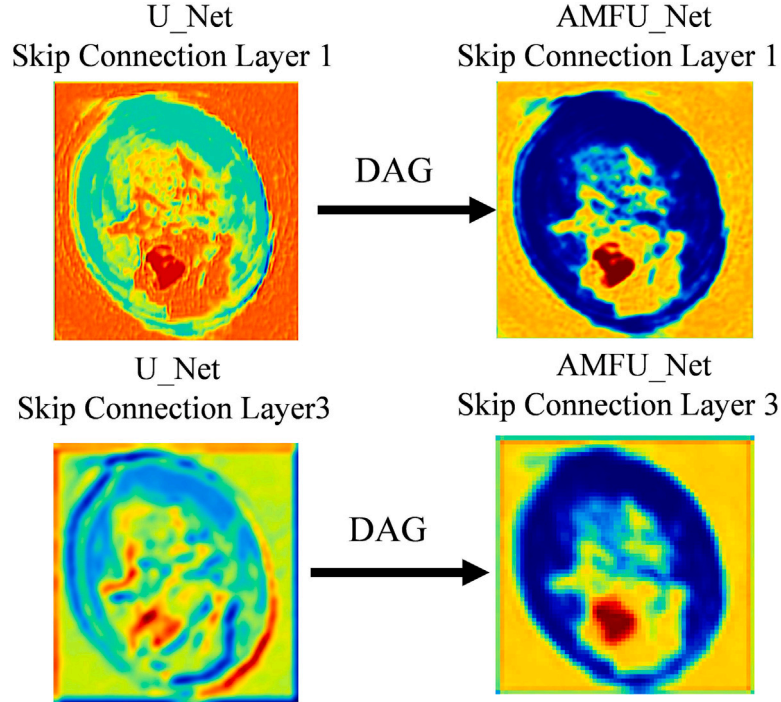


Fig. 12. Comparison of the effect of the DAG module on skip connection attention maps. Compared with the traditional U-Net, the introduction of DAG in AMFU-Net shows activation regions more focused on tumor edges and effective suppression of background interference on skip connections in the first and third layers, suggesting its advantages in feature selection and cross-layer information transfer.

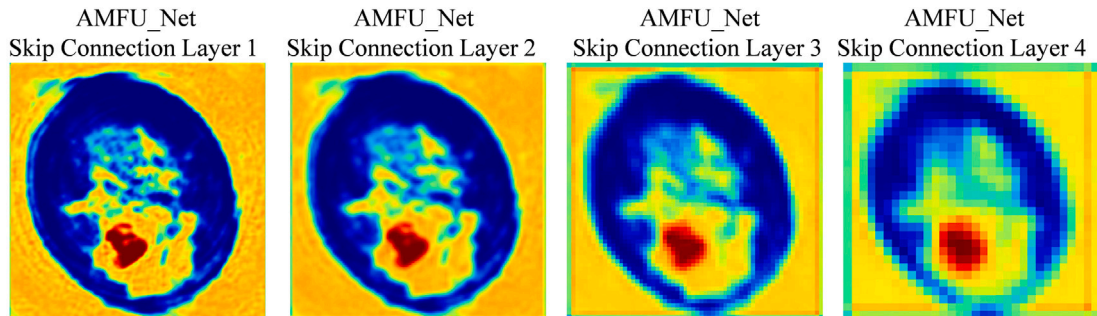


Fig. 13. Attention maps of each skip connection layer of the network after the introduction of the DAG module are shown. It can be observed that the attentional activation remains highly consistent throughout the network and focuses on key regions, which verifies the stability and effectiveness of the DAG module in the process of cross-scale feature fusion.

with the baseline U-Net for the DAG module. In comparison to the baseline U-Net, the DAG-enhanced attention graph exhibits more focused activation around the tumor boundaries while more successfully suppressing irrelevant background information, as seen in Fig. 12. This suggests an improvement in feature selection and information transfer across network layers.

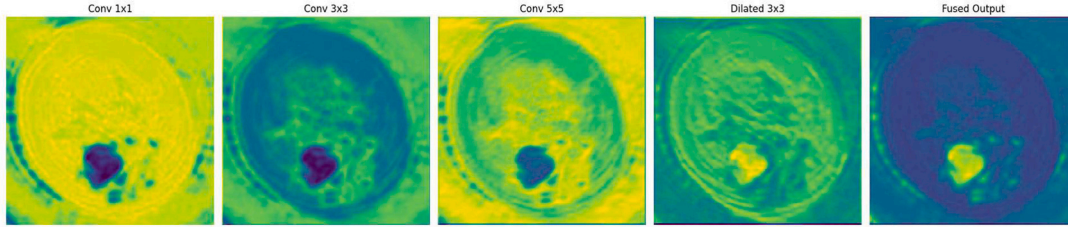
Furthermore, following DAG integration, Fig. 13 shows the attentional mapping across all skip connection layers, illuminating how DAG accomplishes focused and consistent feature selection throughout the network.

To validate the contribution of each component in the MFEF module, we performed ablation experiments by varying the way convolutional branches are combined. Specifically, we tested six architectural variants: the corresponding segmentation performances, measured in terms of the Dice, Jaccard, 95HD, and ASD metrics, are summarized in Table 5. The performance of the MFEF module is summarized in the following table. Among these metrics, the complete configuration ($1 \times 1 + 3 \times 3 + 5 \times 5 + \text{GAP} + \text{dilated } 3 \times 3$) achieved the best results, which confirms the complementary value of the branches in enhancing multiscale feature fusion.

Table 5

Quantitative comparison of different component combinations in the MFEF module.

Configuration	Dice (%)	Jaccard (%)	95HD	ASD
$1 \times 1 + 3 \times 3 + 5 \times 5 + \text{GAP} + \text{dilated } 3 \times 3$	90.19 ± 3.71	82.21 ± 4.16	7.38 ± 0.84	1.74 ± 0.53
$1 \times 1 + 3 \times 3 + 5 \times 5 + \text{dilated } 3 \times 3$	85.14 ± 3.89	77.61 ± 4.87	8.72 ± 0.91	2.54 ± 0.44
$1 \times 1 + 3 \times 3 + 5 \times 5 + \text{GAP}$	87 ± 3.65	77.96 ± 4.73	9.27 ± 1.24	4.11 ± 1.57
$1 \times 1 + 3 \times 3 + \text{GAP} + \text{dilated } 3 \times 3$	86.81 ± 4.31	77.55 ± 4.29	9.09 ± 1.01	2.62 ± 0.38
$3 \times 3 + 5 \times 5 + \text{GAP} + \text{dilated } 3 \times 3$	87.05 ± 4.22	78 ± 4.75	8.11 ± 0.83	2.34 ± 0.42
$3 \times 3 + \text{GAP} + \text{dilated } 3 \times 3$	82.97 ± 4.31	73.27 ± 5.93	10.12 ± 1.12	2.91 ± 0.47

**Fig. 14.** Visualization of the feature maps from each branch in the MFEF module.

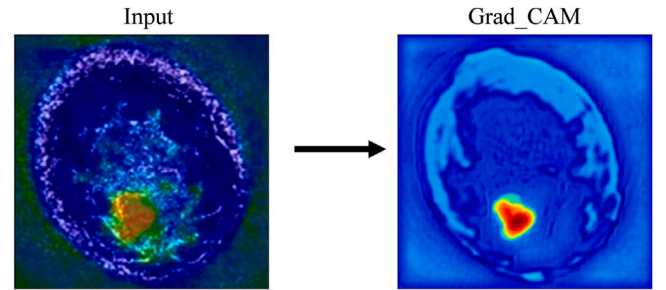
To further investigate the multi-scale behavior of the MFEF module, we visualized the feature maps produced by each convolutional branch, as illustrated in Fig. 14. The 1×1 convolution branch preserves strong intensity contrast within the tumor region, highlighting its core area with minimal background interference. The 3×3 convolution branch captures tumor boundary details and fine textures, aiding in delineating lesion edges. The 5×5 convolution branch emphasizes the overall shape and structure of the tumor, extracting mid-level spatial features while smoothing local variations. The dilated 3×3 convolution, with an enlarged receptive field, captures global context and enhances semantic understanding of the tumor surroundings. The fused output combines these complementary representations, resulting in a clear and complete tumor response with well-defined contours and contextual continuity. These visualizations confirm that MFEF effectively extracts tumor-specific information at multiple scales, supporting both accurate segmentation and interpretable feature enhancement.

In addition, to fully assess the interpretability of AMFU-Net and to validate the impact of feature learning across the network, I employed Gradient Weighted Class Activation Mapping (Grad-CAM). Grad-CAM generates coarse-grained plots highlighting important regions in the image used to predict segmentation masks. Fig. 15 shows the Grad-CAM visualization results of our proposed AMFU-Net.

The Grad-CAM results show that our model learns discriminative features for lesion segmentation and also reduces false positive responses. This improved localization accuracy and focus coincide with the improved quantization performance, further confirming the effectiveness of the DAG and MFEF modules in guiding the network to focus on clinically relevant regions.

5.4. Comparison with state-of-the-art methods

We further compared AMFU-Net with several advanced segmentation architectures, including U-Net [13], Attention U-Net [24], MDA-Net [27], U-Net++ [30], CA-Net [14], and nnU-Net [41]. To ensure the fairness and reproducibility of the experiments, all competing models were retrained from scratch using the same clinical USCT dataset and the identical training/validation/test split (8:1:1) as AMFU-Net: retrained on a unified hardware platform based on the official open source code, trained using the same training/validation/testing segmentation, input image resolution (256×256), and data enhancement strategy (random flipping, rotating, and luminance adjustment). The training process was fixed using the Adam optimizer (initial learning rate of 0.001), and a learning rate scheduler that halves the learning rate every 10 epochs was employed. All models were trained with a batch size of 12 for 200 epochs. As shown in Table 6, AMFU-Net

**Fig. 15.** Grad-CAM visualization of AMFU-Net, highlighting its ability to localize lesion regions accurately while suppressing false positives, thereby demonstrating the interpretability and effectiveness of the DAG and MFEF modules.**Table 6**Performance Comparison of Different Methods (mean $\pm$ standard deviation).

Methods	Dice (%)	Jaccard (%)	95HD	ASD
U-Net [13]	89.06 ± 5.12	80.85 ± 6.12	9.24 ± 1.13	3.03 ± 0.52
Attention-Net [24]	81.88 ± 6.21	73.15 ± 7.48	9.05 ± 1.09	3.09 ± 0.60
MDA-Net [27]	90.44 ± 4.26	83.10 ± 4.63	8.98 ± 2.54	2.72 ± 0.98
UNet++ [30]	82.93 ± 7.45	74.75 ± 6.98	11.09 ± 1.24	3.23 ± 0.61
CA-Net [14]	88.11 ± 4.86	79.42 ± 5.45	8.24 ± 0.97	2.04 ± 0.36
nnU-Net [41]	90.03 ± 3.85	82.62 ± 4.91	7.94 ± 0.91	1.57 ± 0.34
AMFU-Net	$91.42 \pm 3.43^*$	$84.54 \pm 3.98^*$	$5.61 \pm 1.59^*$	$1.36 \pm 0.79^*$

* Statistically significant improvement over U-Net and MDA-Net at $p < 0.05$ level.

achieved superior performance across all metrics. In particular, compared to the highest metrics of other networks, the Dice and Jaccard coefficients improved by 0.98% and 1.44%, respectively, confirming its strong capability in precise breast lesion segmentation.

The segmentation comparison graph is displayed in Fig. 16. Compared to existing segmentation networks, our network is exceptional at effectively segmenting lesions with irregular edge areas and smaller lesions. The main reason AMFU-Net is better than other segmentation networks is because (1) DAG for adaptive feature selection combines global maximum pooling (salient region model) and global average pooling (contextual statistics) to dynamically adjust the encoder, in contrast to single-branch attentional mechanisms (e.g., Attentional U-Net [28]). Eliminate inter-decoder connections. By reducing information loss in the fuzzy boundary region and focusing on the target critical region, this dual-path design increases the accuracy of lesion segmentation. (2) The MFEF, a multi-scale feature extraction and fusion

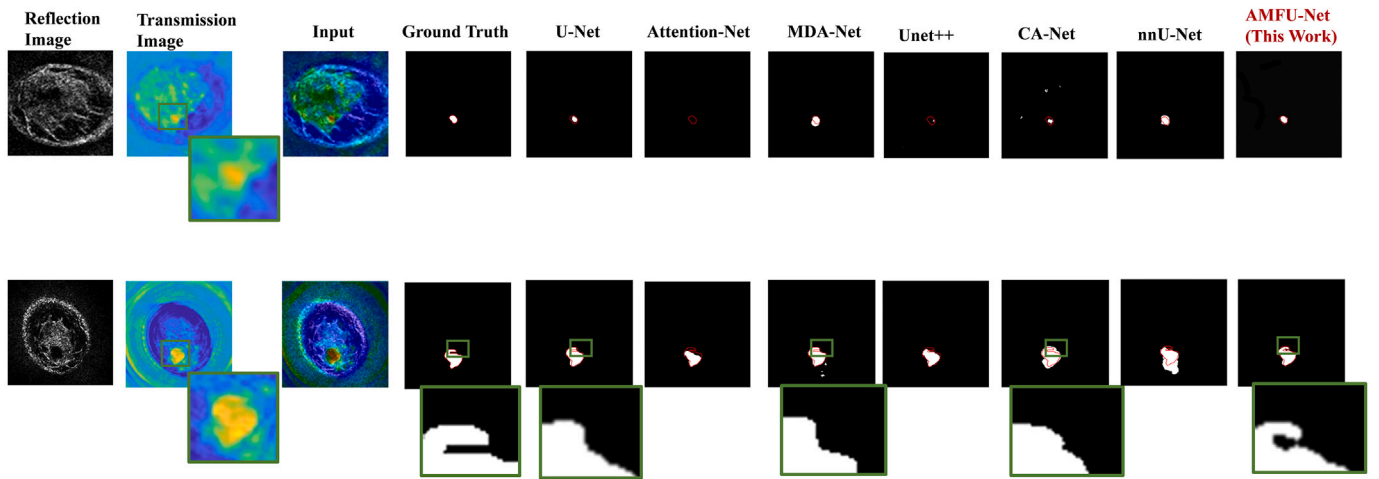


Fig. 16. Comparison of segmentation results of different segmentation networks on clinical data. The red curve is the boundary of the breast lesion. The green border is the zoomed-in area.

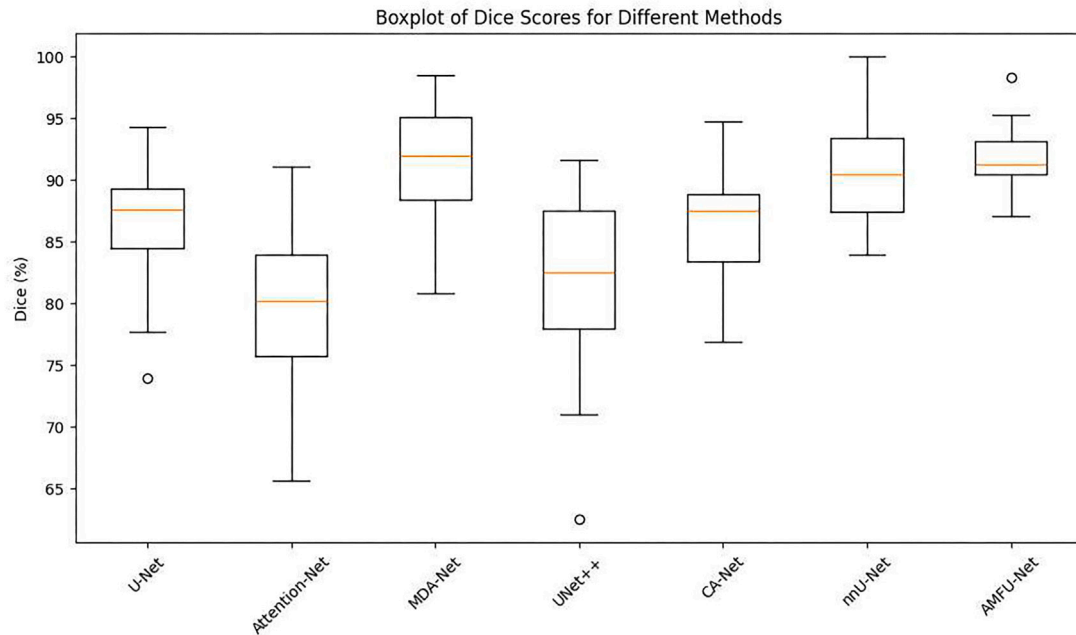


Fig. 17. The distribution of Dice, Jaccard, 95% Hausdorff Distance (95HD), and Average Surface Distance (ASD) for all test samples is shown for each segmentation method. AMFU-Net demonstrates higher median accuracy and lower variance, indicating more stable and robust segmentation performance compared to baseline models.

module with dynamic weighting, uses dilation convolution (rate = 2) and parallel multicore convolution (1×1 , 3×3 , 5×5) to capture varied lesion sizes. Tumors of various sizes and forms can be segmented with the aid of the capacity to extract and choose weights for image features at various scales.

To further evaluate the segmentation performance, we plotted the distribution of Dice scores obtained by different methods on the test set, as shown in Fig. 17. The boxplot compares U-Net, Attention-UNet, MDA-Net, UNet++, CA-Net, nnU-Net, and the proposed AMFU-Net. It can be observed that the proposed AMFU-Net achieves the highest median Dice score and a relatively compact interquartile range, indicating both superior accuracy and more stable performance across cases. In contrast, the conventional U-Net and Attention-UNet exhibit noticeably lower median values and larger variations, while MDA-Net, UNet++, CA-Net, and nnU-Net provide intermediate performance. These results further confirm the effectiveness of the proposed fusion-based network in improving segmentation quality compared with existing methods.

5.5. Changes in segmentation performance with lesion characteristics

In order to deeply investigate the specific effects of different lesion features on the segmentation performance of breast images, this study manually grouped and categorized all 22 samples in the test set in terms of three dimensions, namely, lesion size, contrast, and shape regularity, in order to support the subsequent group comparison experiments.

In terms of lesion size, we quantitatively classified lesion areas based on the number of pixels they occupied in the corresponding mask image. Samples with less than 300 pixels were defined as small tumors, and those with more than 300 pixels were defined as large tumors. The statistical results showed that there were 6 small tumor samples and 16 large tumor samples.

For the color input image, this study first converted the original RGB color image to a single-channel grayscale image based on a standard luminance weighting formula (Eq. (27)) to obtain the grayscale value (I) of each pixel; then the grayscale contrast between the lesion area and

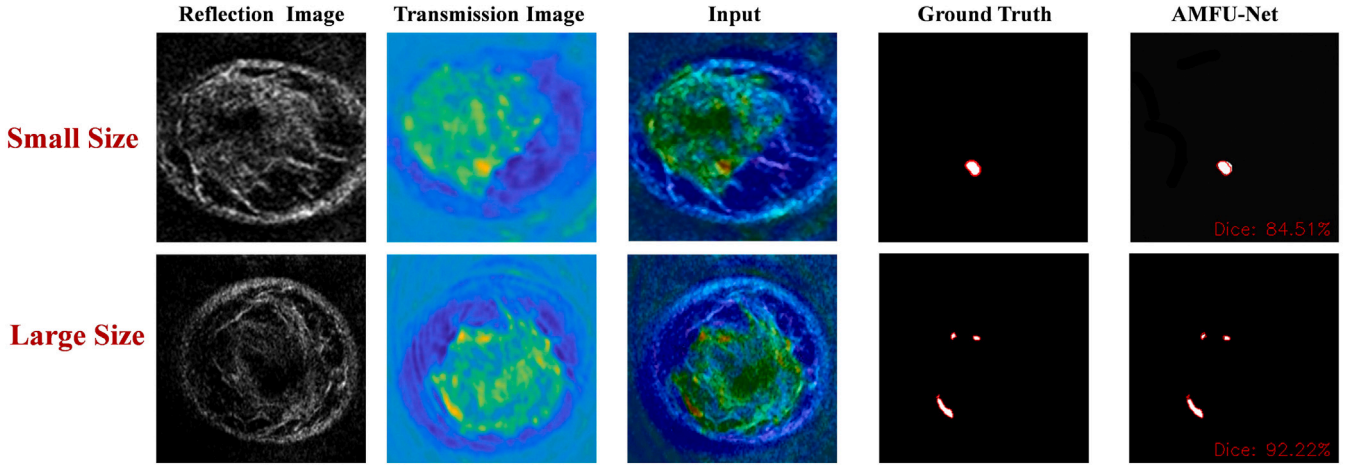


Fig. 18. The model demonstrates higher segmentation accuracy on large tumor samples (Dice: 92.22%) compared to small tumors (Dice: 85.51%), indicating improved stability when processing larger lesion areas.

the background area was evaluated using Eq. (28): the absolute value of the difference between the average grayscale values of the lesion area and the background area was calculated as the contrast index. Finally, the samples were categorized according to a threshold ($C < 50$ for low contrast and $C \geq 50$ for high contrast). Application of this method to analyze the results showed the existence of 4 low-contrast samples and 18 high-contrast samples.

$$I = 0.299 * R + 0.587 * G + 0.114 * B \quad (27)$$

$$\text{Contrast} = \left| \frac{1}{N_{\text{lesion}}} \sum_{i \in \Omega_{\text{lesion}}} I_i - \frac{1}{N_{\text{background}}} \sum_{j \in \Omega_{\text{background}}} I_j \right| \quad (28)$$

where I denotes the gray value of the image Ω_{lesion} and $\Omega_{\text{background}}$ denote the lesion and background regions, respectively, and N is the number of pixels in the corresponding region.

For the regularity of the shape of the lesion, the Circularity index was used, which ranges from 0 to 1, with larger values indicating a closer approximation to an ideal circle. The formula is shown in the Eq. (29)

$$\text{Circularity} = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2} \quad (29)$$

where **Area** denotes the area of the lesion region (in pixel quantity), **Perimeter** denotes the perimeter of the lesion region (length of the pixel boundary).

In this study, the threshold of circularity was set at 0.7, which was used as a differentiation criterion: samples with a circularity greater than or equal to 0.7 were categorized as regular shapes, and those with an irregular shape were categorized as irregular shapes. The statistics show that there are 10 samples of regular shapes and 12 samples of irregular shapes.

In the specific test, representative small and large tumor samples are selected for comparison, and the results show that the Dice coefficient of small tumor samples is 84.51%, while that of large tumor samples is 92.22%, which indicates that the model has superior segmentation stability when dealing with larger lesions (see Fig. 18).

We extracted different contrast versions based on the same image in order to eliminate the interference of tumor size and shape factors in the contrast analysis. The high-contrast image includes the full fusion and optimization steps, while the low-contrast image is the intermediate result of only completing the first stage of fusion, which excludes the image enhancement process. According to the segmentation results, the low-contrast samples had an accuracy of 84.67%, while the high-contrast samples had a Dice coefficient of 92.22%. This suggests that greater contrast can improve the model's capacity to identify the lesion's boundary, which in turn improves segmentation performance.

Table 7

Segmentation performance under different training proportions.

Training proportion	Training slices (n)	Dice (%)	Jaccard (%)	95HD	ASD
20%	43	74.86 ± 6.84	59.84 ± 7.55	14.72 ± 2.95	3.57 ± 1.28
40%	86	81.27 ± 5.92	68.45 ± 6.65	11.63 ± 2.55	2.71 ± 1.07
60%	129	87.92 ± 4.28	78.46 ± 5.05	8.34 ± 1.98	1.93 ± 0.92
70%	151	90.83 ± 3.62	83.13 ± 4.35	6.58 ± 1.72	1.53 ± 0.83
80%	172	91.42 ± 3.43	84.54 ± 3.98	5.61 ± 1.59	1.36 ± 0.79

In terms of shape regularity, the Dice coefficient of the regular shape samples is 92.22%, while that of the irregular shape samples is 91.71%. Although the difference is relatively small, it still reflects that the model has some advantages in dealing with lesions with more regular geometries (see Fig. 20).

In summary, it can be concluded that visual attributes such as lesion size, contrast, and shape have different degrees of influence on the segmentation performance of the model; especially under the conditions of larger lesion area or higher gray contrast, the model shows better segmentation accuracy and robustness.

5.6. Effect of training set size

To investigate the effect of dataset scale, we conducted a controlled data-scaling study under a fixed 8:1:1 split. The validation set (10%) and test set (10%) were kept unchanged, and only the training size was varied by randomly sub-sampling the training pool. Specifically, the number of training slices was increased from 20% to 80% of the total 215 slices (i.e., 43, 86, 129, 151, and 172 slices). As shown in Table 7 and Fig. 21, the Dice score increases rapidly when the training data grows from 40 slices to 150 slices, and the performance gain becomes marginal afterwards, indicating a saturation trend. With the full training size of 172 slices (80%), the proposed AMFU-Net achieves 91.42% Dice on the fixed test set.

5.7. Computational cost and real-time feasibility analysis

All training and evaluation procedures were implemented using the PyTorch deep learning framework. All timing metrics were measured on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), an Intel Core i9 CPU, and 64 GB of RAM. To address concerns regarding the system's practicality for real-time clinical applications, we quantitatively evaluated the computational overhead of both the fusion and segmentation processes.

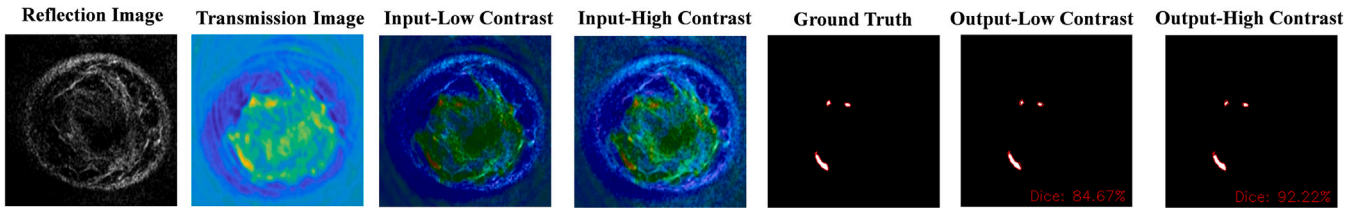


Fig. 19. The segmentation accuracy significantly improves in high-contrast images (Dice: 84.67%) compared to low-contrast ones (Dice: 92.22%), revealing the model's sensitivity to boundary visibility enhanced by contrast.

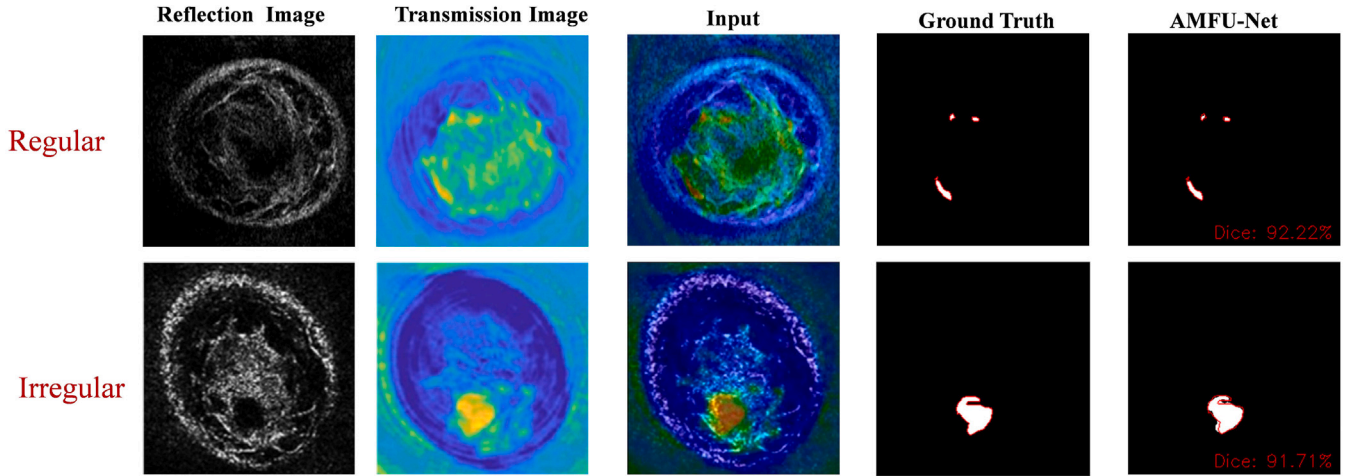


Fig. 20. The Dice scores for regular (92.22%) and irregular (91.71%) shapes are close; the model shows slightly better performance on lesions with regular geometries.

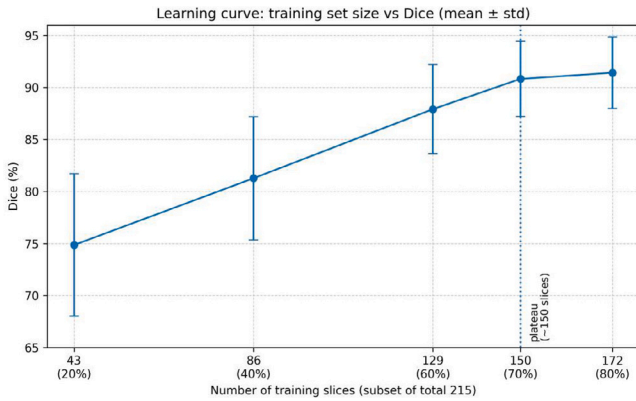


Fig. 21. With fixed validation/test sets, increasing training slices (43 → 172) yields rapid Dice improvements up to 150 slices, followed by diminishing gains and reduced variability, suggesting performance saturation and stabilization.

The PSO-based image fusion module was tested over 100 randomly selected 2D slices (size: 256×256), achieving an average runtime of 1.21 ± 0.12 s for 100 optimization iterations per slice. The average inference time of the AMFU-Net model was 0.15 ± 0.03 s per slice. When performed sequentially, the total processing time per 2D image amounts to approximately 1.37 s.

For a typical 3D breast USCT data volume containing 50 slices, the end-to-end processing time (fusion + segmentation) is approximately 1 min, which is acceptable for near real-time or batch-mode clinical workflows. These results demonstrate that the proposed framework is computationally efficient and viable for integration into practical imaging pipelines.

6. Discussions

We combined transmission and reflection images using an image fusion technique that was optimized by the PSO algorithm and HIS-based discrete wavelet transform. The results demonstrated that the information entropy of the images was higher than the original imaging results, and the fused images preserved the structural and parametric fidelity of the two, thus improving the image quality. However, PSO's iterative optimization may result in higher computational overhead; hence, further research into the algorithm's computational overhead issue is necessary.

By adding an MFEF module for multi-scale feature extraction and fusion in the U-net neck and a dual-branch attention gate in the skip connections, our segmentation technique outperforms the U-net. Ablation experiments were carried out to assess each module's performance. The ablation experiment results demonstrate that AMFU-net's components can successfully enhance segmentation performance.

When compared to the original U-net network structure, experimental results comparing this method to others demonstrate that the introduction of dual-branch effectively selects the information from the encoder to the decoder. The MFEF module enhances segmentation accuracy by fusing multi-scale features through adaptive weight allocation. Our method still has significant advantages over other approaches, despite its shortcomings, which include unclear edge contour and inadequate segmentation accuracy.

There are still certain issues that require correction, even though AMFU-Net's outcomes are superior to those of other networks. (1) The boundary module was increased; the findings show that our network's accuracy in boundary delineation for lesions is still lacking. (2) Additional simplification of the module. Both the training parameters and the module's complexity are somewhat increased by the parallel branching that was added to the MFEF module. Future work will focus on reducing computational complexity through lightweight attention

mechanisms and integrating boundary refinement modules to further improve edge accuracy. (3) AMFU-Net still has limitations for low-contrast lesions. Low-contrast lesions: As shown in Fig. 19, lesions with very low contrast to the surrounding tissues are sometimes missed or segmented incompletely. This is likely due to insufficient intensity differences, which reduces the efficacy of convolution and attention-based feature extraction. Complex background interference: In the presence of dense glandular structures or imaging artifacts, the network occasionally misdiagnosed non-lesion regions as tumors. (4) Another important factor affecting model performance in this study is the limited size and diversity of the available dataset. Due to privacy concerns and the challenges associated with acquiring annotated medical images, our dataset comprises a relatively small number of samples collected from a single clinical center, using a single imaging device. The dataset also exhibits imbalanced lesion sizes and variations in image quality. These limitations not only constrain the generalization ability of the proposed model but also significantly hinder the training stability of more complex architectures, particularly transformer-based networks such as TransUNet and SwinUNet. Despite efforts to apply data augmentation and regularization techniques, these models demonstrated noticeable overfitting and unstable convergence behavior during training. To address these challenges, future work will focus on expanding the dataset by incorporating multi-institutional data sources and generating synthetic variations using advanced augmentation or generative methods (e.g., GANs). This will enable a more comprehensive and fair evaluation of different segmentation frameworks, especially those requiring large-scale, diverse data for effective learning. Moreover, to improve the model's applicability and robustness across different clinical scenarios, we also plan to introduce cross-center validation and explore domain generalization or domain adaptation techniques. These strategies aim to enhance the model's stability and performance when applied to unseen imaging settings or patient populations with varied imaging parameters and data distributions.

7. Conclusion

This study presents an integrated framework that synergizes multimodal image fusion with attention-driven deep segmentation to enhance breast ultrasound tomosynthesis. The proposed fusion strategy, which combines HIS discrete wavelet transform with particle swarm optimization, effectively integrates complementary information from reflection and transmission images, preserving structural clarity while enhancing edge definition. Comparative experiments confirm that the fused images yield significantly better segmentation performance than using either modality alone, validating the necessity of the fusion step.

To achieve precise lesion delineation, we developed AMFU-Net, an attention-guided multi-scale network equipped with a dual-branch attention gate (DAG) in skip connections and a multi-scale feature extraction and fusion (MFEF) module at the bottleneck. These components collaboratively enable adaptive feature refinement and cross-scale contextual integration. Quantitative evaluation on a clinical dataset of 215 tomographic slices demonstrates that our method achieves a Dice coefficient of 91.42%, outperforming several state-of-the-art segmentation networks. Ablation studies further verify the individual contributions of both the fusion mechanism and the network modules.

Despite these advancements, the framework still faces challenges in segmenting low-contrast lesions and handling complex backgrounds, and its generalization is limited by the current single-center dataset. Future work will focus on extending the model's robustness through multi-institutional data collection, integrating lightweight attention mechanisms to reduce computational overhead, and incorporating boundary refinement modules to further improve edge accuracy.

In conclusion, this work provides a comprehensive and scalable solution for breast lesion analysis, demonstrating the significant potential of combining multimodal fusion with attention-aware deep learning for clinical ultrasound imaging and beyond.

CRediT authorship contribution statement

Lijuan Chen: Writing – original draft, Software, Methodology. **Dan Gao:** Validation, Investigation, Data curation. **Guojun Zhang:** Supervision, Formal analysis. **Renxin Wang:** Supervision, Formal analysis. **Changde He:** Supervision, Formal analysis. **Xingling Shao:** Supervision, Formal analysis, Conceptualization. **Wendong Zhang:** Supervision, Funding acquisition. **Licheng Jia:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of the AMFU-Net: A Multi-Scale Attention Network for Lesion Segmentation in Fused Breast Ultrasound Imaging.

Acknowledgments

This work was supported in part by the National Key Research and Development Project of China under Grant 2022YFB3204302, in part by the National Natural Science Foundation of China under Grant 61927807, 62320106011 and 62304208, and in part by the Shanxi Province Science Foundation for Youths under Grant 202203021222025.

Data availability

The authors do not have permission to share data.

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